

A PROJECT TITLE

ON

Anomaly-Based Intrusion Detection System Using DBSCAN and Isolation Forest Algorithms

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

Name Samrat Reddy roll 2520030131

Name Nihal Metuku roll 2520030124

Under the Esteemed guidance of

Dr. Shaik Asif

Assistant Professor, CSE

COMPUTER SCIENCE & ENGINEERING



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Name of Title: Anomaly-Based Intrusion Detection System Using DBSCAN and Isolation Forest Algorithms

1. Abstract:

Intrusion Detection Systems (IDS) play a crucial role in protecting computer networks from cyberattacks and unauthorized access. Traditional signature-based detection methods are effective against known attacks but often fail to identify new or evolving threats. This project proposes an anomaly-based intrusion detection system that utilizes two machine learning algorithms, DBSCAN (Density-Based Spatial Clustering of Applications with Noise) and Isolation Forest, to detect abnormal network behavior without relying on predefined attack signatures.

The proposed system first preprocesses network traffic data by handling missing values, removing irrelevant attributes, encoding categorical features, and normalizing numerical data. DBSCAN is then used to group similar network activities based on density while identifying sparse data points as potential anomalies. Isolation Forest complements this approach by isolating anomalous observations through randomly generated decision trees, making it highly efficient for large datasets. The outputs of both algorithms are analyzed and compared to improve the reliability of anomaly detection.

The system is evaluated using publicly available network intrusion datasets such as NSL-KDD or CIC-IDS2017. Performance is measured using metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis. Visualization techniques are also employed to demonstrate the distribution of normal and anomalous traffic.

The proposed approach aims to improve the detection of previously unseen attacks while reducing false alarms. By combining clustering-based and isolation-based anomaly detection techniques, the system provides a scalable and efficient solution for modern cybersecurity applications. The project demonstrates how unsupervised and semi-supervised machine learning methods can enhance network security and assist organizations in identifying suspicious activities at an early stage. This work highlights the growing importance of artificial intelligence in developing intelligent, adaptive, and real-time cybersecurity solutions.

2. INTRODUCTION:

With the rapid growth of digital communication and internet-based services, cyber threats have become increasingly sophisticated. Organizations require intelligent security systems capable of detecting both known and unknown attacks. Intrusion Detection Systems (IDS) monitor network traffic and identify suspicious activities that may compromise system security.

Traditional signature-based IDS depend on predefined attack patterns, making them ineffective against zero-day attacks and evolving threats. Anomaly-based detection overcomes this limitation by learning the characteristics of normal network behavior and identifying deviations as potential intrusions.

This project implements an anomaly detection system using DBSCAN and Isolation Forest. DBSCAN identifies clusters of normal traffic while treating isolated data points as anomalies. Isolation Forest detects anomalies by isolating unusual observations through random partitioning. Combining these techniques provides a more robust mechanism for identifying suspicious network activities and improving cybersecurity.

3. SOFTWARE REQUIREMENTS:

CATEGORY	REQUIREMNET
Operating System	Windows 10/11 or Ubuntu Linux
Programming Language	Python 3.10+
IDE	Visual Studio Code / Jupyter Notebook
Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Dataset	NSL-KDD or CIC-IDS2017
Version Control	Git & GitHub
Visualization	Matplotlib, Seaborn
Hardware	Minimum Intel i5/Ryzen 5, 8 GB RAM, 10 GB Free Storage



Signature of the Guide



Course Instructor